

Code

```
In[*]:= (* Brute force construction of all Graphs on n vertices up to Isomorphism*)
allEdges[n_] := Flatten[Table[UndirectedEdge[i, j], {i, 1, n}, {j, i + 1, n}]];
allConnected[n_] := Select[Map[Graph[Range[n], #, VertexLabels → Automatic] &,
    Subsets[allEdges[n]]], ConnectedGraphQ];
allConnectedUpToIso[n_] := DeleteDuplicates[allConnected[n], IsomorphicGraphQ];
(*Helper Functions*)
PerfectDistanceQ[G_] := Union[Flatten[Round[GraphDistanceMatrix[G]]]] ==
    Range[0, Binomial[VertexCount[G], 2]]
WeightsList[G_] :=
    Flatten[Permutations[#] & /@ Select[Subsets[Range[Binomial[VertexCount[G], 2]],
        {EdgeCount[G]}], MemberQ[#, 1] && MemberQ[#, 2] &], 1]

(*Construct Graph Sets*)
vertex6 = allConnectedUpToIso[6];
(*vertex6=allConnectedUpToIso[6];*)
```

```

In[*]:= (*
  vertex5 should take around a second
  vertex6 took around 8 minutes on my laptop
*)
(*CHOOSE SET OF GRAPHS TO TEST*)
GraphList = vertex6;

MinimalPerfectDistanceGraphs = {};
NotPerfectDistanceGraphs = {};

AbsoluteTiming[
  While[Length[GraphList] > 0,
    G = First[GraphList];
    GraphList = Delete[GraphList, 1];

    (* Test weights until a PDG is found or weights exhausted*)
    (*PerfectDistanceGraph =
      Null if G does not support a perfect distance weighting*)
    WL = WeightsList[G];

    PerfectDistanceGraph =
      Catch[Do[If[PerfectDistanceQ[Graph[EdgeList[G], EdgeWeight → WL[[j]]]],
        Throw[G]], {j, 1, Length[WL]}]];

    (*Update MPDG and NPD: Comparisons to Null are probelmatic*)
    If[Head[PerfectDistanceGraph] == Graph,
      AppendTo[MinimalPerfectDistanceGraphs, PerfectDistanceGraph]];
    If[PerfectDistanceGraph == Null, AppendTo[NotPerfectDistanceGraphs, G]];

    (*Trim Graph List*)
    GraphList =
      Select[GraphList, Not[IsomorphicSubgraphQ[PerfectDistanceGraph, #]] &];
  ];

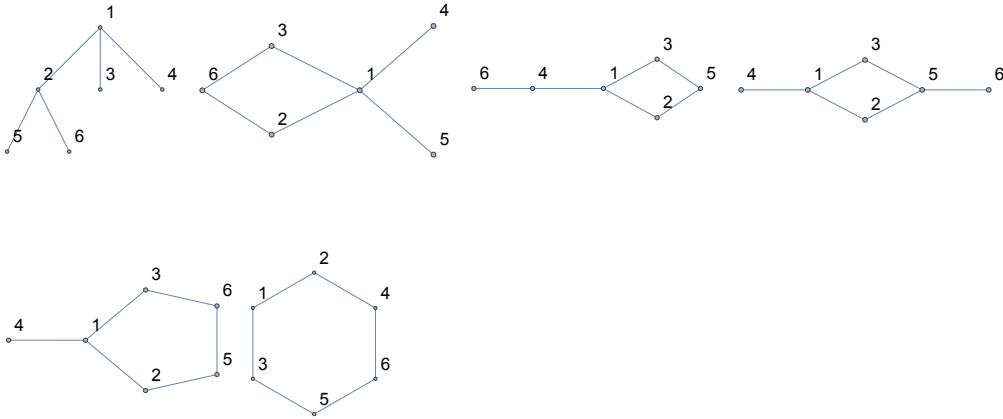
Print["Minimal Perfect Distance Graphs: "]
TableForm[MinimalPerfectDistanceGraphs, TableDirections → Row]

Print["Not Perfect Distance Graphs: "]
TableForm[NotPerfectDistanceGraphs, TableDirections → Row]

Out[*]=
{387.503, Null}

```

Minimal Perfect Distance Graphs:



Not Perfect Distance Graphs:

